A POINT OF A BLOWING UP LIES IN THE AFFINE CHART OF A LOCAL EQUATION EXACTLY WHEN THAT EQUATION CUTS OUT THE EXCEPTIONAL DIVISOR THERE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Statement span 8469-8478 and proof span 8480-8503 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof uses the author's notation for the blowup algebra and writes a displayed map from the affine chart's coordinate ring into the local ring at the point; the display is preserved unchanged.

Editorial note. 2 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The affine description of a blowing up by blowup algebras, with the exceptional divisor over the affine chart cut out by the image of the chosen element. Used at proof lines 8480-8483 and 8496-8498. Gives one direction immediately and supplies the description of the blowup algebra used to define the comparison map.
- The universal property of blowing up, cited from the same chapter. Cited at proof lines 8501-8503. Upgrades the ring homomorphism built from the local divisibility into a morphism over the blowing up, which exhibits the point inside the affine chart.

Source reference. Original definition 01OG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2193–2193. Same-chapter reference used in the retained author text at source line(s) 8496. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0806: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10171–10171. Same-chapter reference used in the

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retained author text at source line(s) 8500, 8513, 8607, 8615, 8816. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $b : X' \to X$ be the blowing up of the scheme X along a closed subscheme Z. Let $U = \mathrm{Spec}(A)$ be an affine open of X and let $I \subset A$ be the ideal corresponding to $Z \cap U$. Let $a \in I$ and let $x' \in X'$ be a point mapping to a point of U. Then x' is a point of the affine open $U' = \mathrm{Spec}(A[\frac{I}{a}])$ if and only if the image of a in $\mathcal{O}_{X',x'}$ cuts out the exceptional divisor.*

Proof. Since the exceptional divisor over U' is cut out by the image of a in $A' = A[\frac{I}{a}]$ one direction is clear. Conversely, assume that the image of a in $\mathcal{O}_{X',x'}$ cuts out E. Since every element of I maps to an element of the ideal defining E over $b^{-1}(U)$ we see that elements of I become divisible by a in $\mathcal{O}_{X',x'}$. Thus for $f \in I^n$ we can write $f = \psi(f)a^n$ for some $\psi(f) \in \mathcal{O}_{X',x'}$. Observe that since a maps to a nonzerodivisor of $\mathcal{O}_{X',x'}$ the element $\psi(f)$ is uniquely characterized by this. Then we define

$$A' \longrightarrow \mathcal{O}_{X',x'}, \quad f/a^n \longmapsto \psi(f)$$

Here we use the description of blowup algebras given following Algebra, Definition 01OG. The uniqueness mentioned above shows that this is an A-algebra homomorphism. This gives a morphism $\mathrm{Spec}(\mathcal{O}_{X',x"}) \to \mathrm{Spec}(A') = U'$. By the universal property of blowing up (Lemma 0806) this is a morphism over X', which of course implies that $x' \in U'$. □